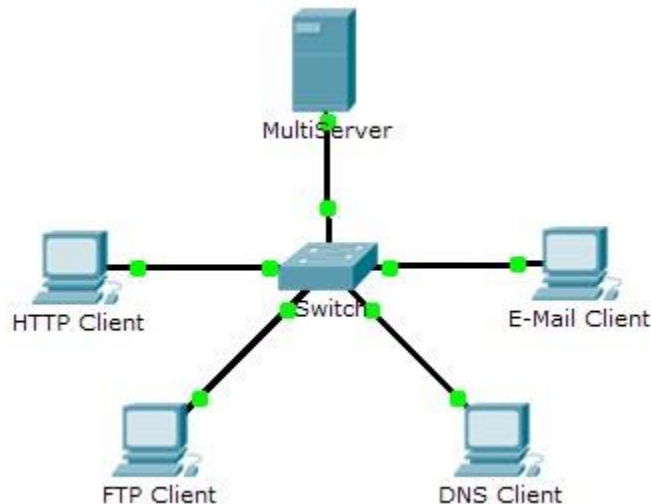


Packet Tracer Simulation - TCP and UDP Communications

Topology



Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding the TCP and UDP in detail. Simulation mode provides the ability to view the functionality of the different protocols.

As data moves through the network, it is broken down into smaller pieces and identified in some fashion so that the pieces can be put back together. Each of these pieces is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer. Packet Tracer Simulation mode enables the user to view each of the protocols and the associated PDU. The steps outlined below lead the user through the process of requesting services using various applications available on a client PC.

This activity provides an opportunity to explore the functionality of the TCP and UDP protocols, multiplexing and the function of port numbers in determining which local application requested the data or is sending the data.

Part 1: Generate Network Traffic in Simulation Mode

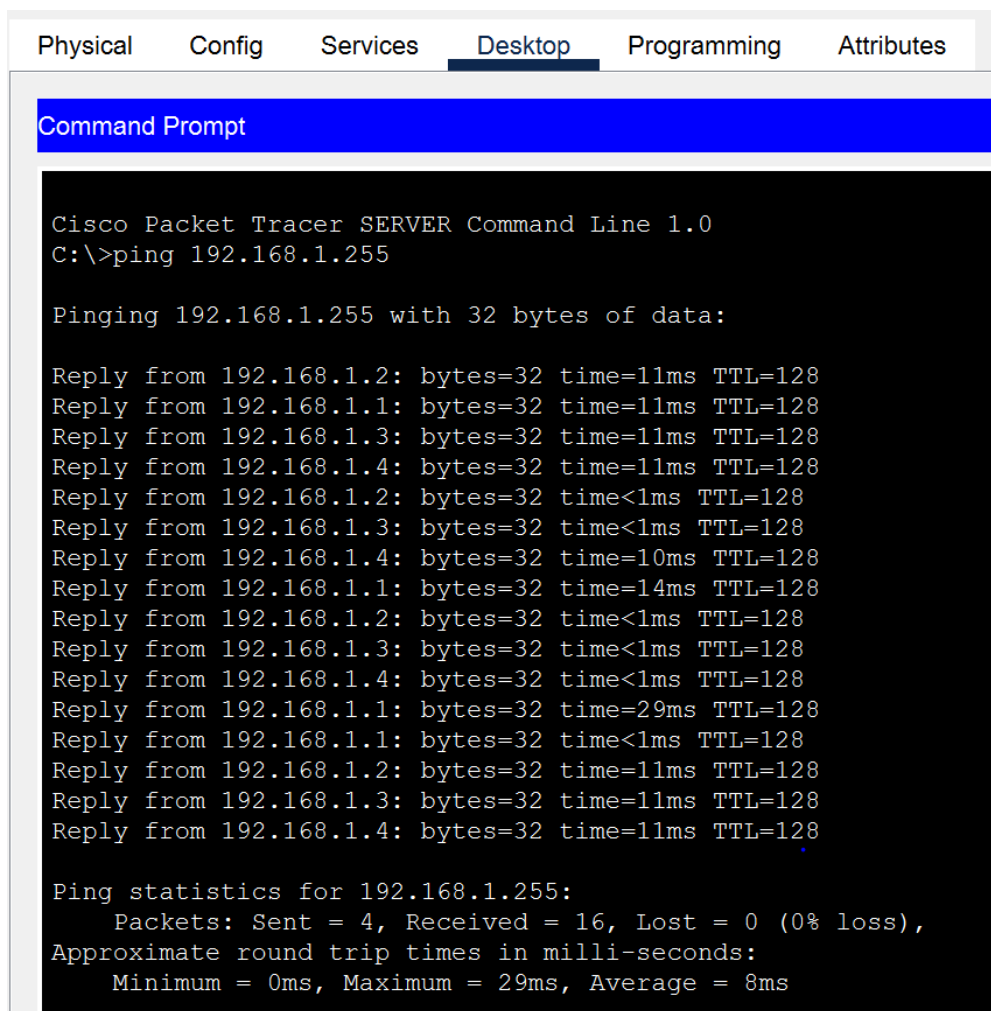
Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following tasks task to reduce the amount of network traffic viewed in the simulation. a.

Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **ping 192.168.1.255** command. This will take a few seconds as every device on the network responds to **MultiServer**.

- c. Close the **MultiServer** window.



The screenshot shows the Cisco Packet Tracer interface with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the following text:

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.1.255

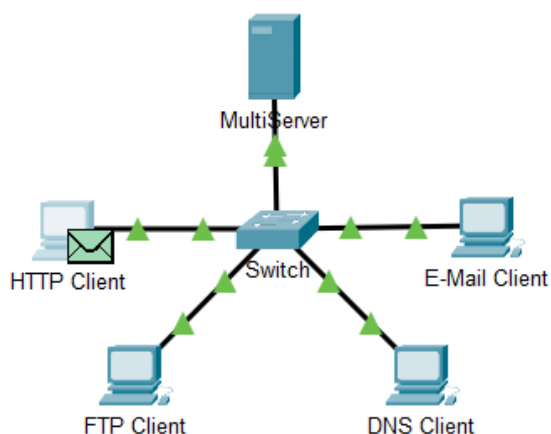
Pinging 192.168.1.255 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=11ms TTL=128
Reply from 192.168.1.1: bytes=32 time=11ms TTL=128
Reply from 192.168.1.3: bytes=32 time=11ms TTL=128
Reply from 192.168.1.4: bytes=32 time=11ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=10ms TTL=128
Reply from 192.168.1.1: bytes=32 time=14ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=29ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=11ms TTL=128
Reply from 192.168.1.3: bytes=32 time=11ms TTL=128
Reply from 192.168.1.4: bytes=32 time=11ms TTL=128

Ping statistics for 192.168.1.255:
    Packets: Sent = 4, Received = 16, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 29ms, Average = 8ms
```

Step 2: Generate web (HTTP) traffic.

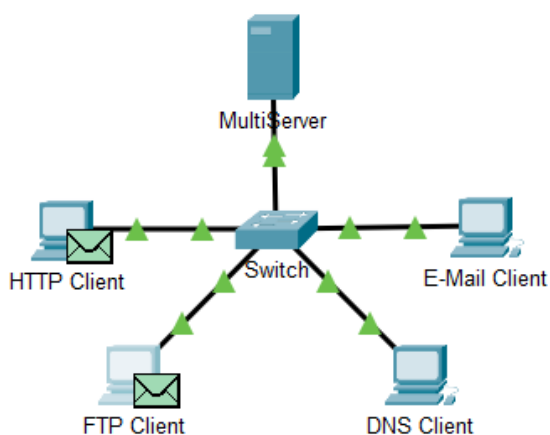
- Switch to Simulation mode.
- Click **HTTP Client** and click the **Desktop** tab > **Web Browser**.
- In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the simulation window.
- Minimize, but do not close, the **HTTP Client** configuration window.



Event List		
Vis.	Time(sec)	Last Device
	0.000	--

Step 3: Generate FTP traffic.

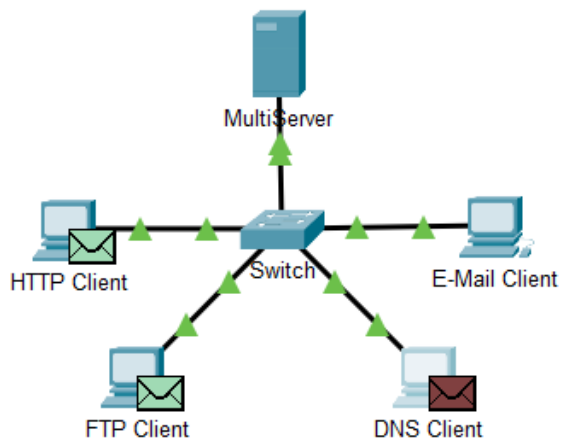
- Click **FTP Client** and click the **Desktop** tab > **Command Prompt**.
- Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.
- Minimize, but do not close, the **FTP Client** configuration window.



Event List		
Vis.	Time(sec)	Last Device
	0.000	--
	0.000	--

Step 4: Generate DNS traffic.

- Click **DNS Client** and click the **Desktop** tab > **Command Prompt**.
- Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.
- Minimize, but do not close, the **DNS Client** configuration window.

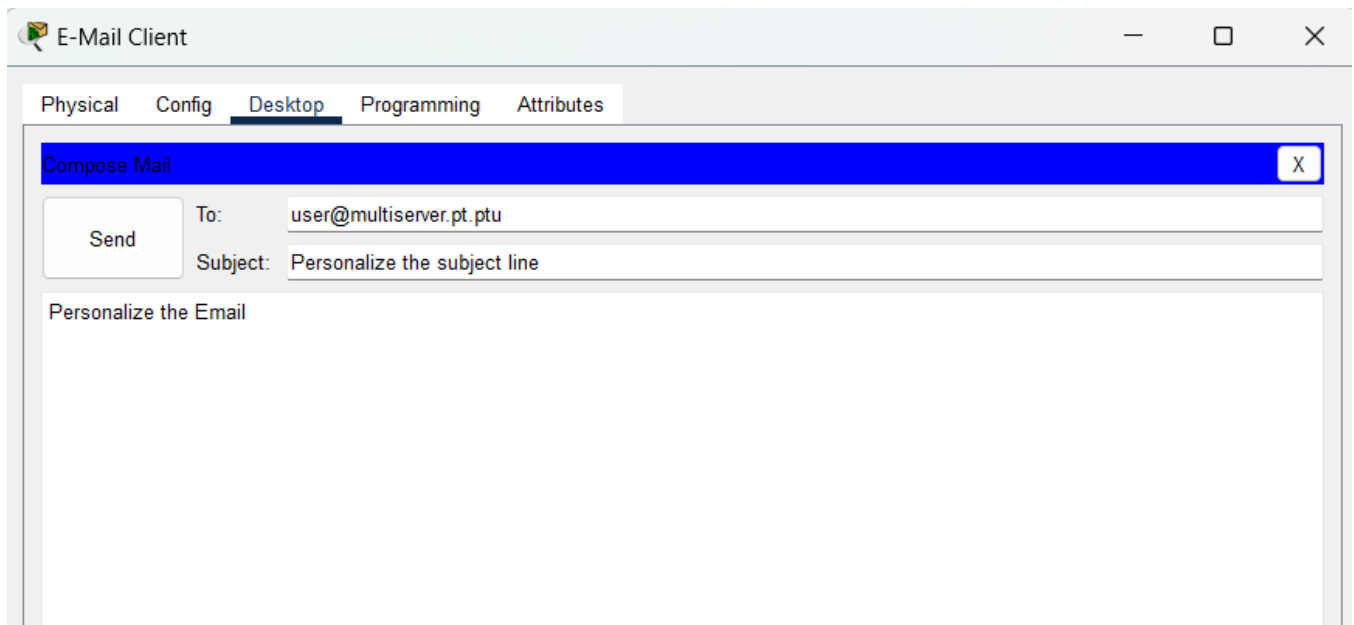


Event List

Vis.	Time(sec)	Last Device
	0.000	--
	0.000	--
	0.000	--

Step 5: Generate Email traffic.

- Click **E-Mail Client** and click the **Desktop** tab > **E Mail** tool.
- Click **Compose** and enter the following information:
 - To:** user@multiserver.pt.ptu
 - Subject:** Personalize the subject line
 - E-Mail Body:** Personalize the Email
- Click **Send**.
- Minimize, but do not close, the **E-Mail Client** configuration window.



Sending mail to user@multiserver.pt.ptu , with subject : Personalize the subject line .. Mail Server: 192.168.1.254

Cancel
Send/Receive

Step 6: Verify that the traffic is generated and ready for simulation.

Every client computer should have PDUs listed in the Simulation Panel.

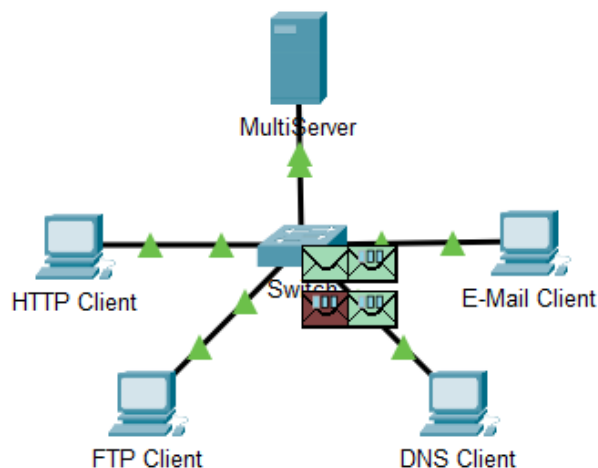
Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	HTTP Client	TCP
	0.000	--	FTP Client	TCP
	0.000	--	DNS Client	DNS
	0.000	--	E-Mail Client	TCP

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine multiplexing as all of the traffic crosses the network.

You will now use the **Capture/Forward** button and the **Back** button in the Simulation Panel.

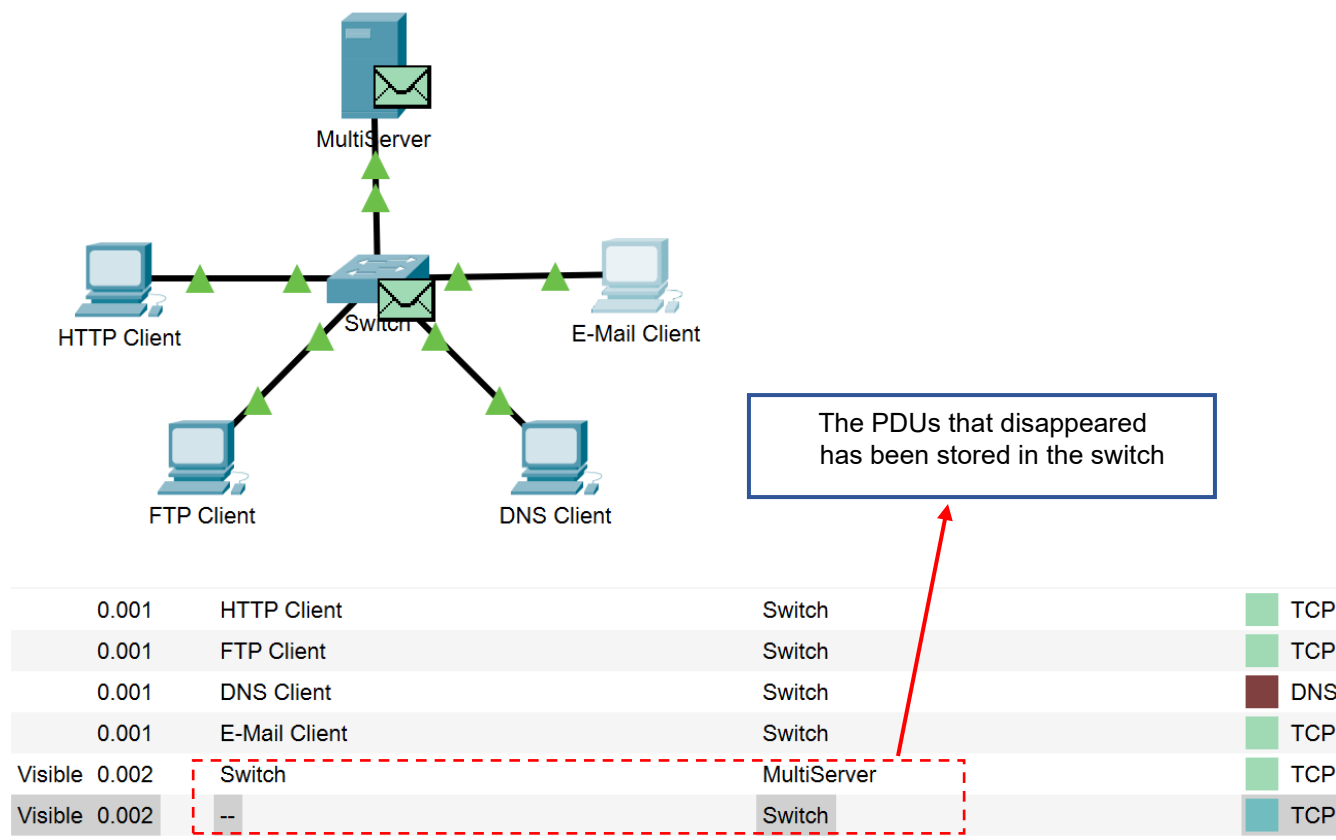
a. Click **Capture/Forward** once. All of the PDUs are transferred to the switch.



All PDUs are transferred to the switch

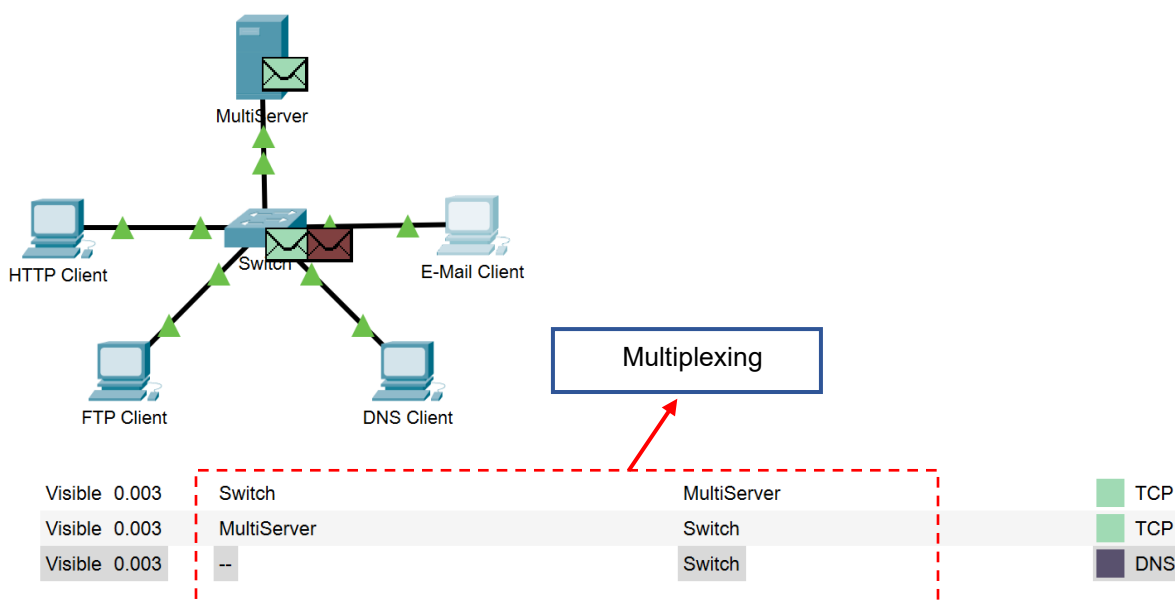
Visible	0.001	HTTP Client	Switch	TCP
Visible	0.001	FTP Client	Switch	TCP
Visible	0.001	DNS Client	Switch	DNS
Visible	0.001	E-Mail Client	Switch	TCP

- b. Click **Capture/Forward** again. Some of the PDUs disappear. What do you think happened to them?

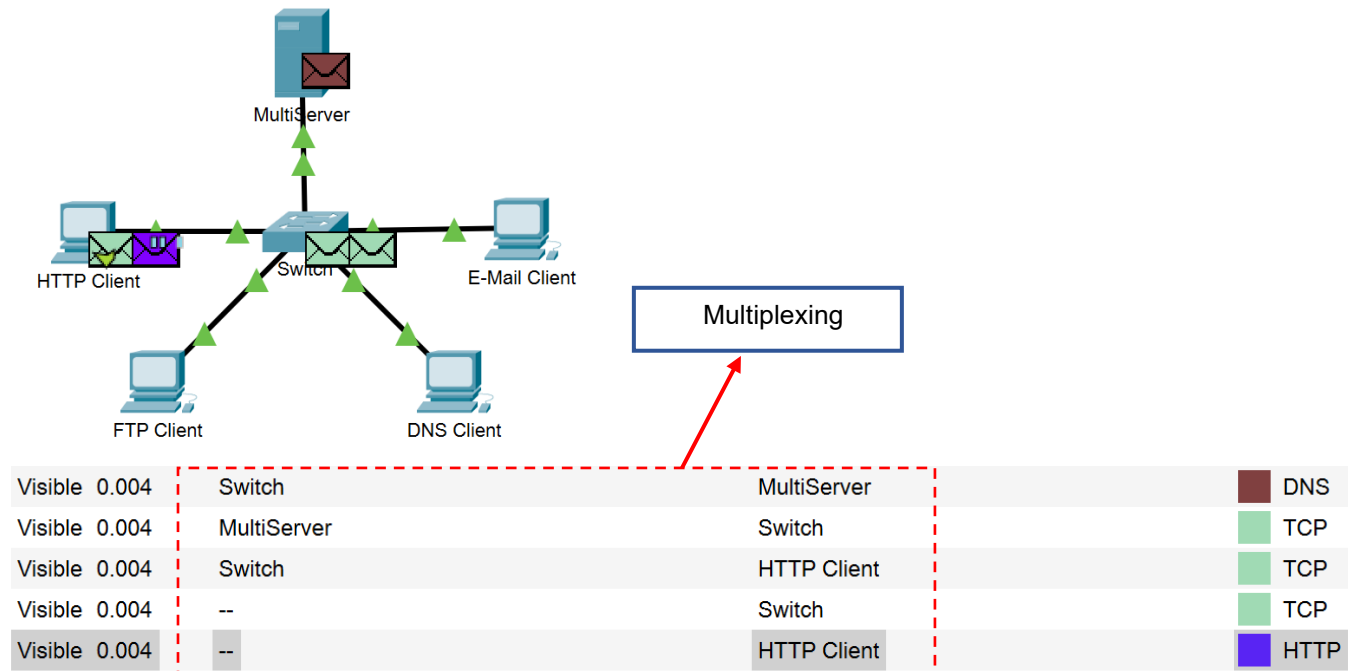


- c. Click **Capture/Forward** six times. All clients should have received a reply. Note that only one PDU can cross a wire in each direction at any given time. What is this called?

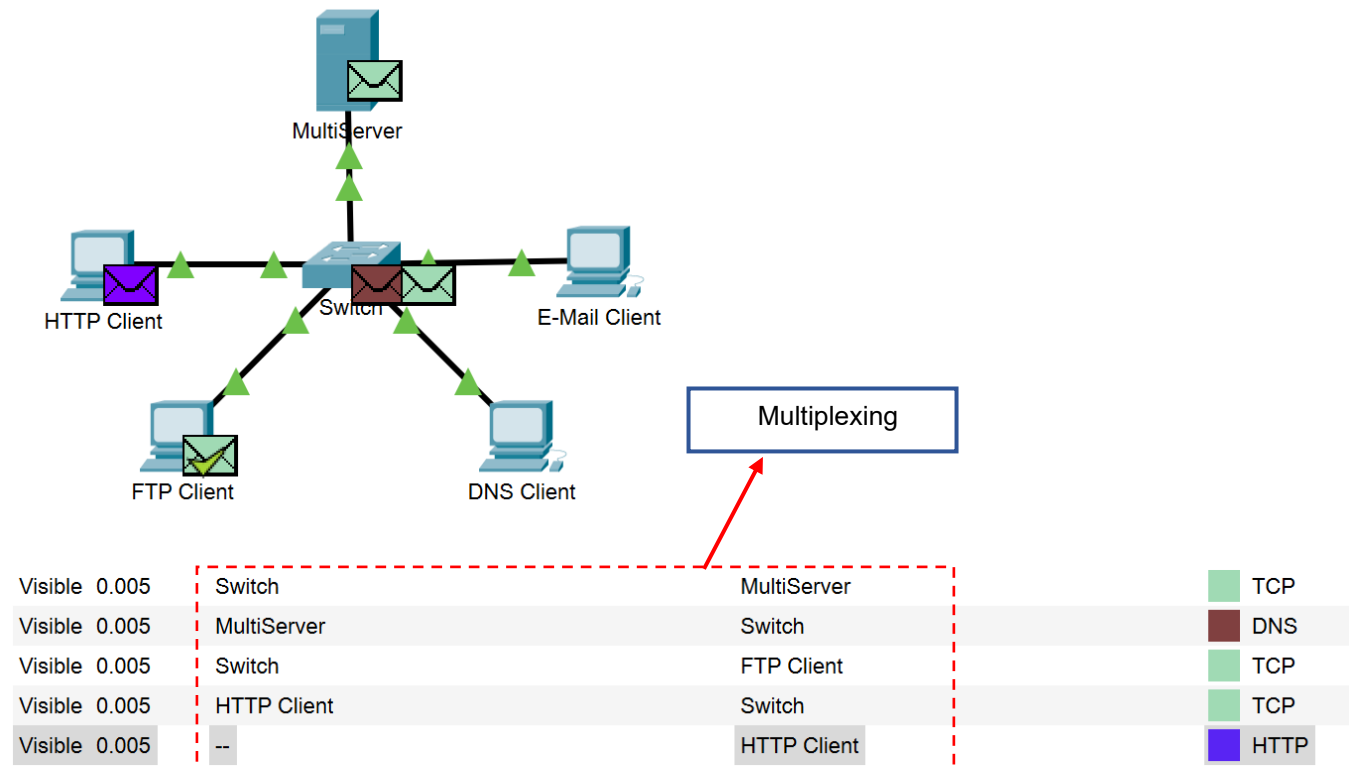
First time:



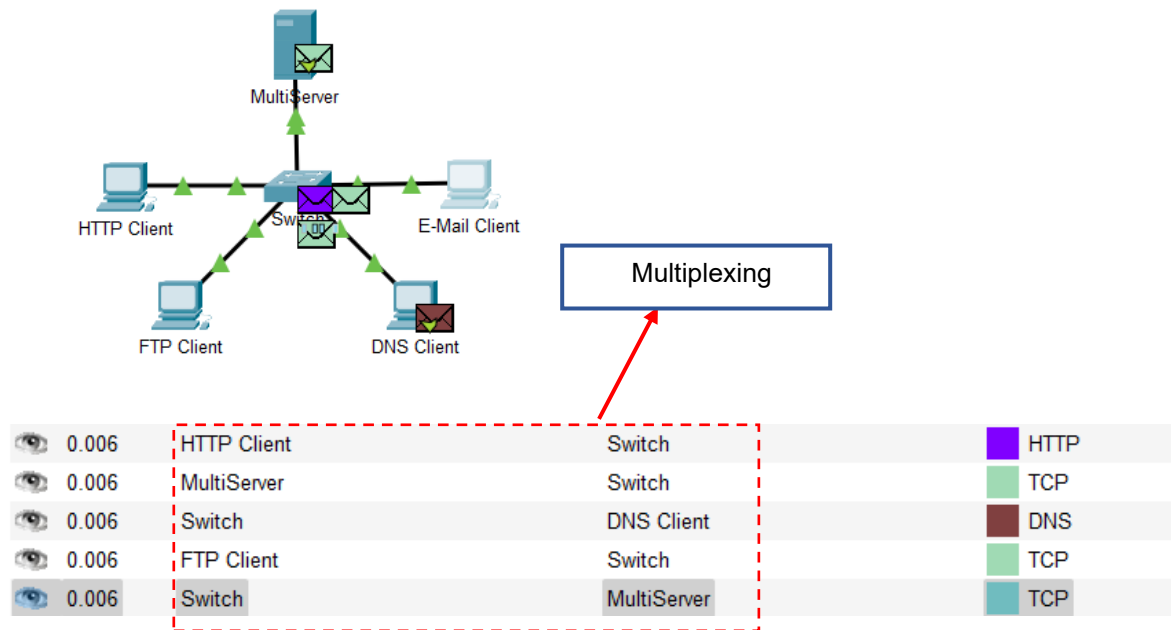
Second time:



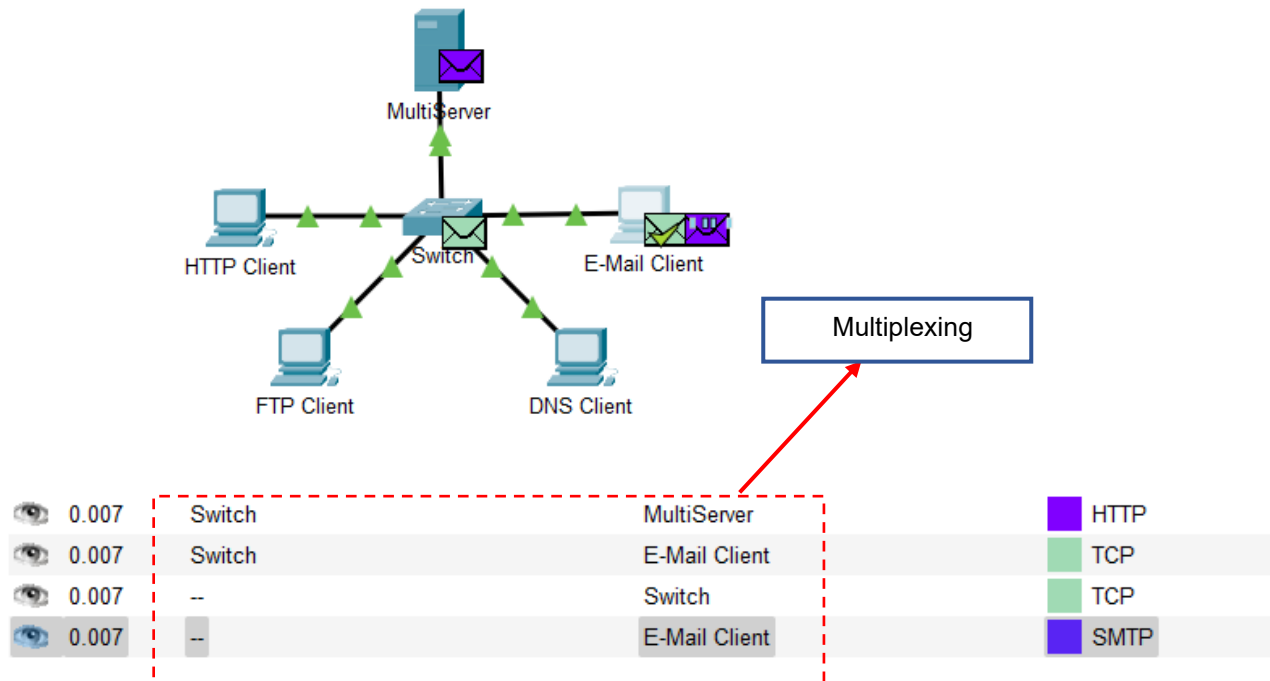
Third time:



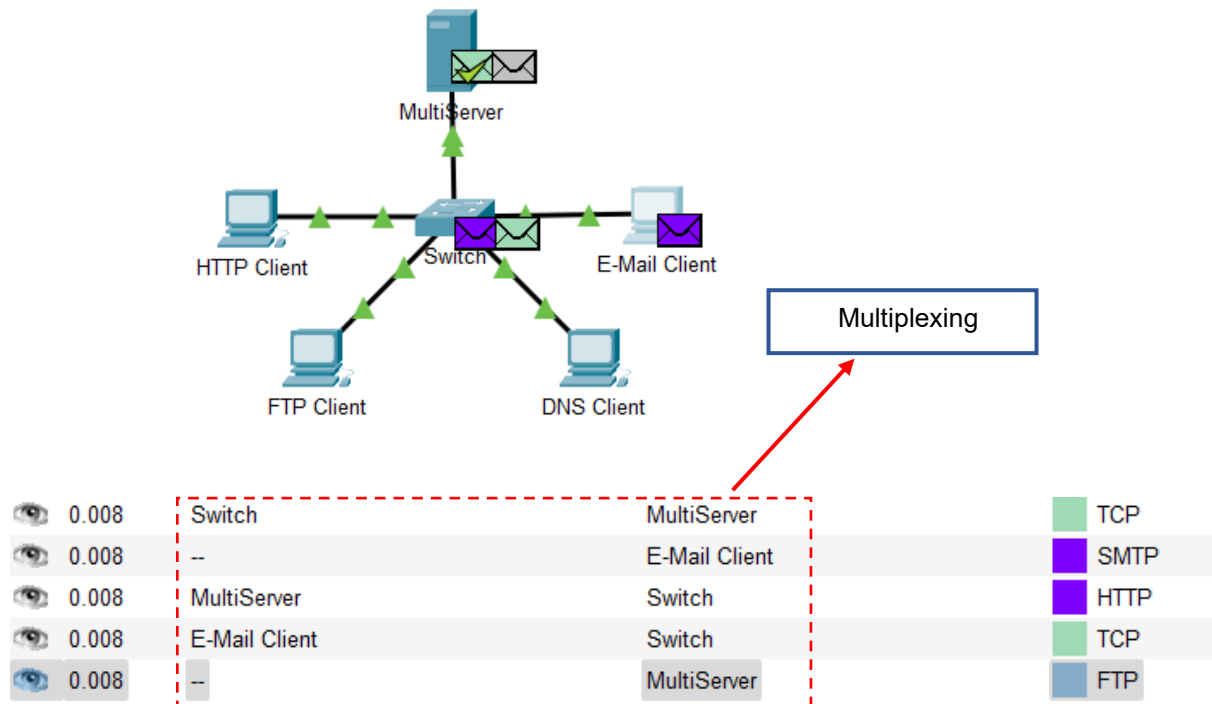
Fourth time:



Fifth Time:



Sixth Time:



- d. A variety of PDUs appears in the event list in the upper right pane of the simulation window. Why are they so many different colors?



- e. Click **Back** eight times. This should reset the simulation.

Note: Do not click **Reset Simulation** any time during this activity; if you do, you will need to repeat the steps in Part 1.

Step 2: Examine HTTP traffic as the clients communicate with the server.

- Filter the traffic that is currently displayed to display only **HTTP** and **TCP** PDUs filter the traffic that is currently displayed:
 - Click **Edit Filters** and toggle the **Show All/None** check box.
 - Select **HTTP** and **TCP**. Click anywhere outside of the Edit Filters box to hide it. The Visible Events should now display only **HTTP** and **TCP** PDUs.

Packet Tracer Simulation - TCP and UDP Communications

Simulation Panel

Event List

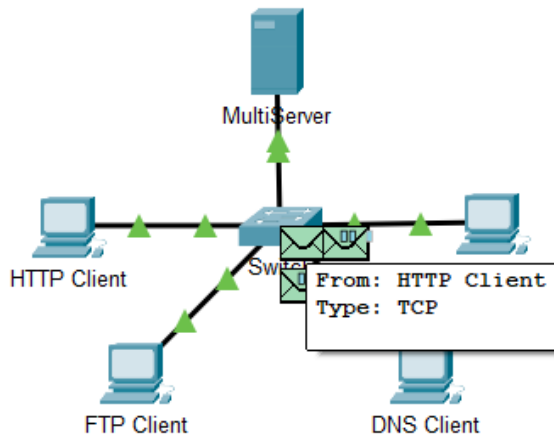
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	HTTP Client	TCP
	0.000	--	FTP Client	TCP
	0.000	--	E-Mail Client	TCP
	0.001	HTTP Client	Switch	TCP
	0.001	FTP Client	Switch	TCP
	0.001	E-Mail Client	Switch	TCP
	0.002	Switch	MultiServer	TCP
	0.002	--	Switch	TCP
	0.003	Switch	MultiServer	TCP
	0.003	MultiServer	Switch	TCP
	0.004	MultiServer	Switch	TCP
	0.004	Switch	HTTP Client	TCP
	0.004	--	Switch	TCP
	0.004	--	HTTP Client	HTTP
	0.005	Switch	MultiServer	TCP
	0.005	Switch	FTP Client	TCP
	0.005	HTTP Client	Switch	TCP
	0.005	--	HTTP Client	HTTP

Cisco Packet Tracer

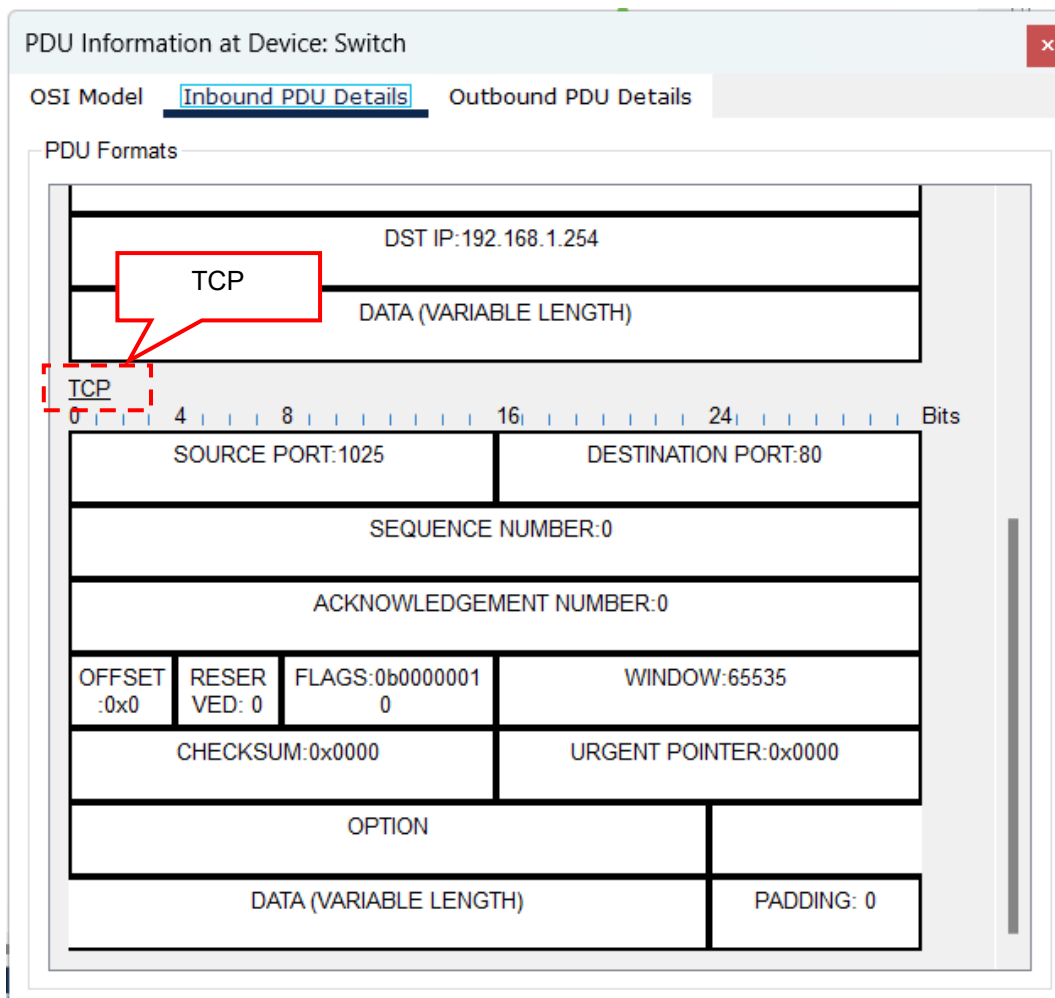
IPv4 IPv6 Misc

Bluetooth CAPWAP CDP
 DTP EAPOL FTP
 H.323 HTTP HTTPS
 IPSec ISAKMP IoT
 IoT TCP LACP LLDP
 Meraki NETFLOW NTP
 PAP POP3 PPP
 PPPoE PTP RADIUS
 REP RTP SCCP
 SMTP SNMP SSH
 STP SYSLOG TACACS+
 TCP TFTP Telnet
 UDP USB VTP

- b. Click **Capture/Forward**. Hold your mouse above each PDU until you find one that originates from **HTTP Client**. Click the PDU envelope to open it.



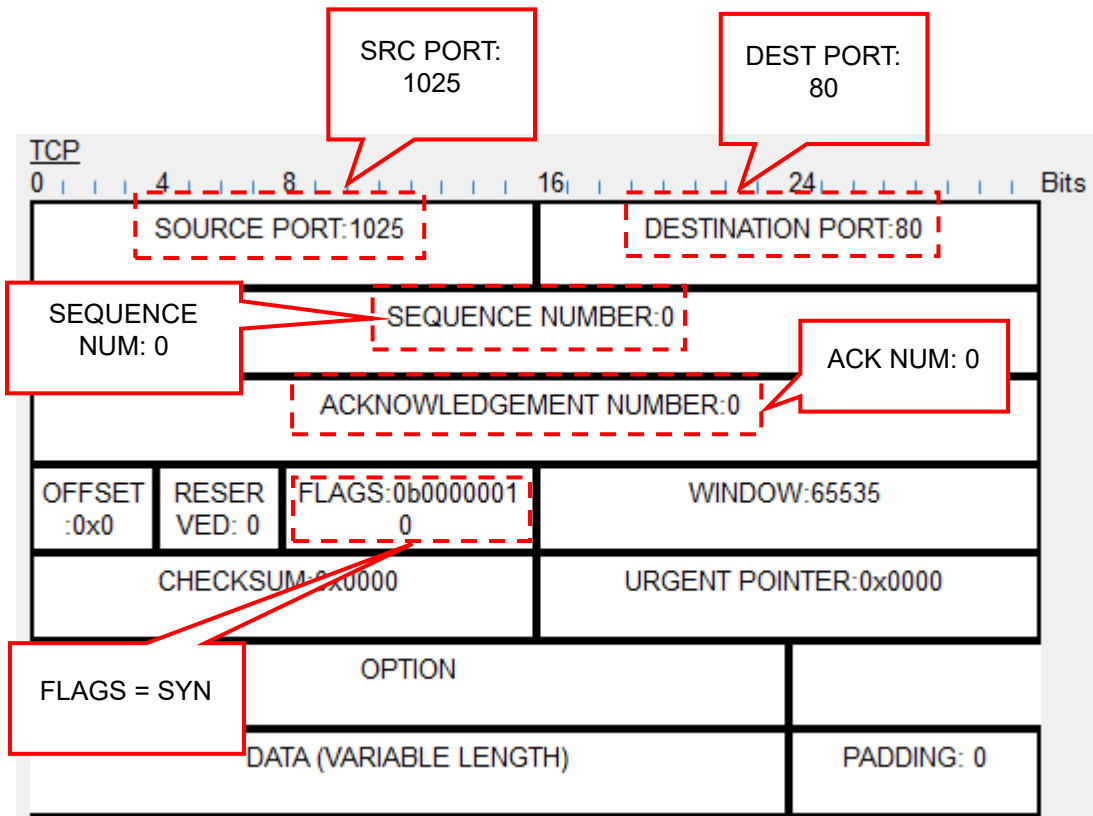
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?



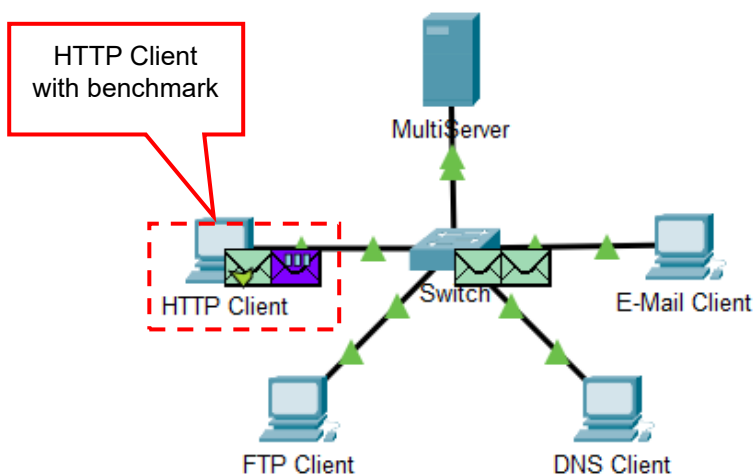
Are these communications considered to be reliable?

Yes. TCP is reliable transport which has a connection oriented between sending and receiving process.

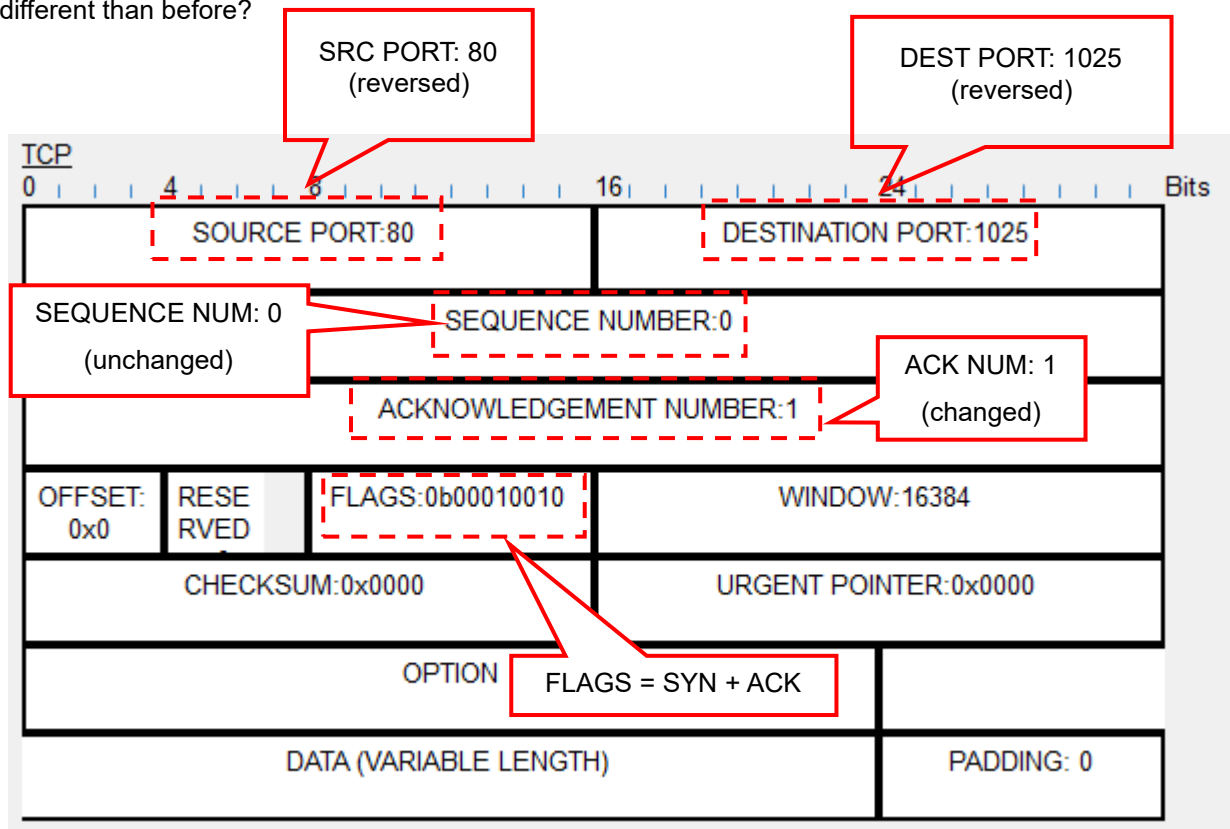
- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field? \



- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **HTTP Client** with a checkmark.

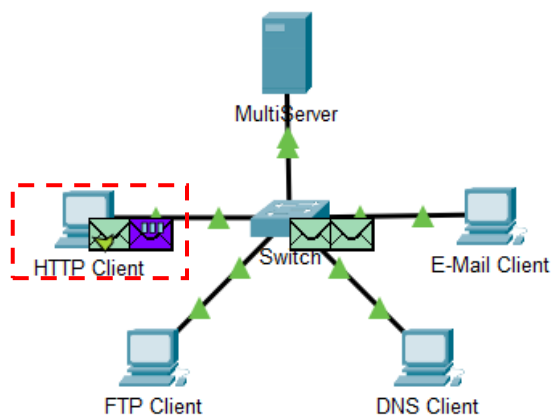


- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

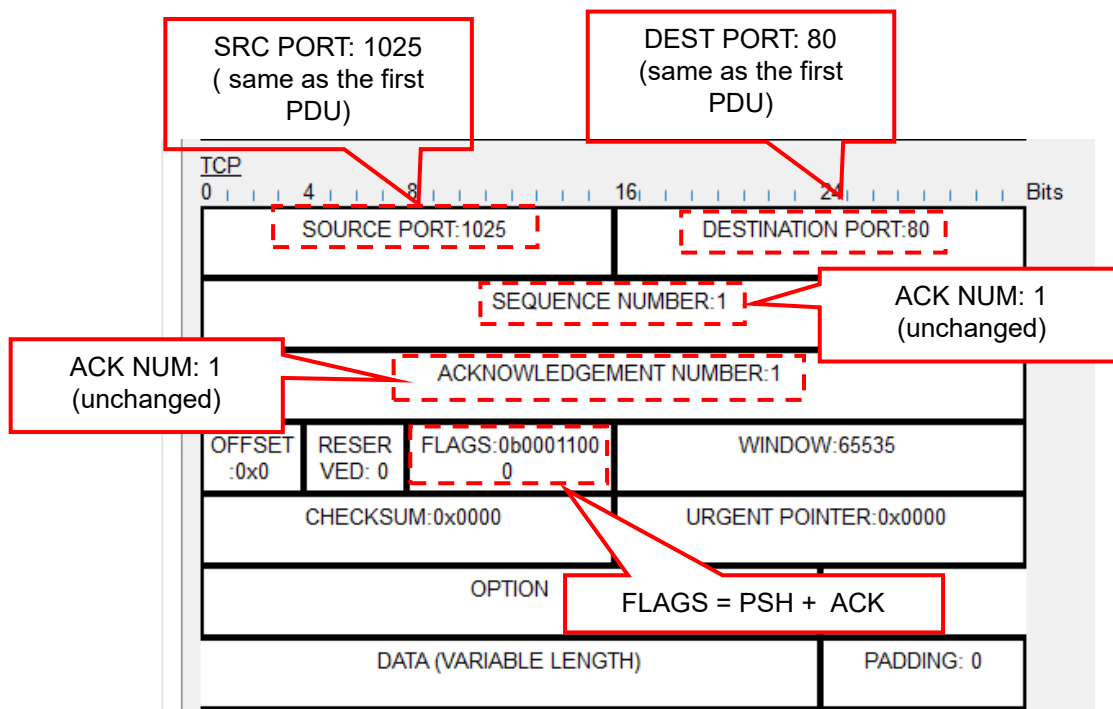


The SRC PORT and DEST PORT are reversed, ACK NUM changed to 1 and FLAGS changed to SYN + ACK

- g. There is a second PDU of a different color, which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.



- h. What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

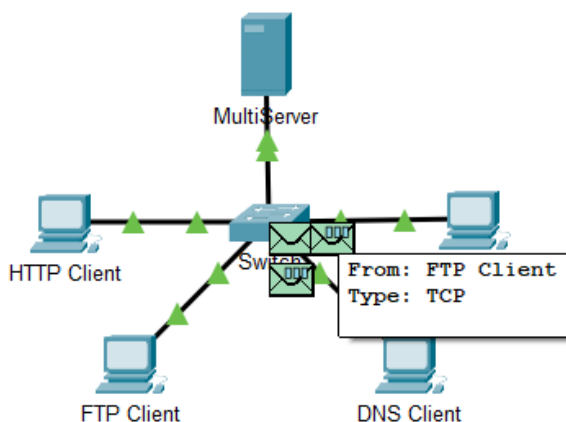


SRC PORT and DEST PORT are reversed, SEQUENCE NUM and ACK NUM changed to 1 and FLAGS changed to PSH + ACK

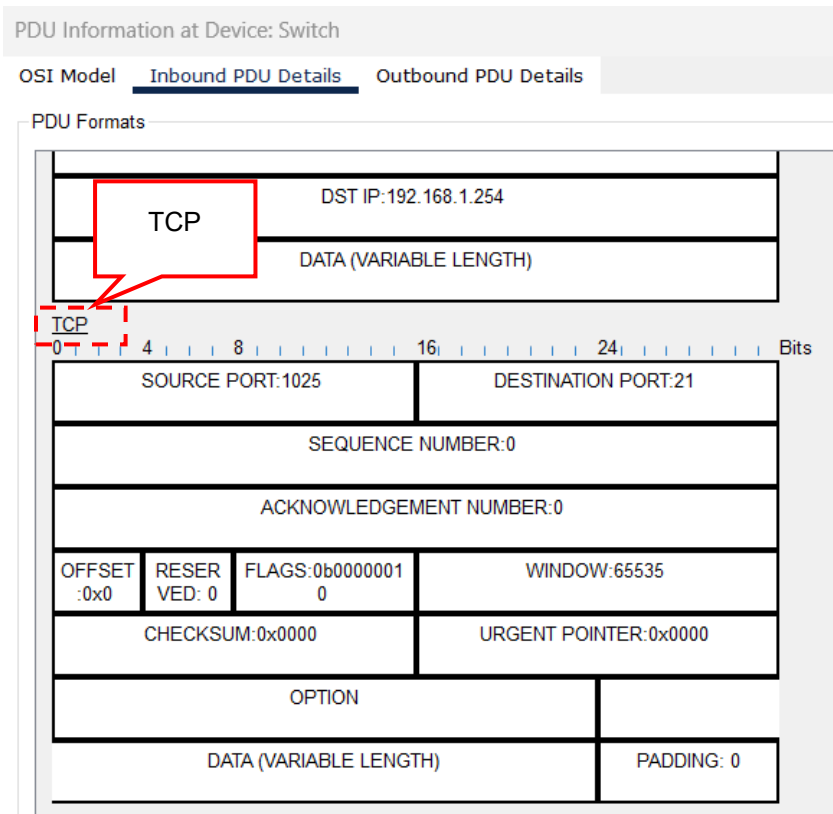
- i. Click **Back** until the simulation is reset.

Step 3: Examine FTP traffic as the clients communicate with the server.

- In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.
- Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **FTP Client**. Click that PDU envelope to open it.



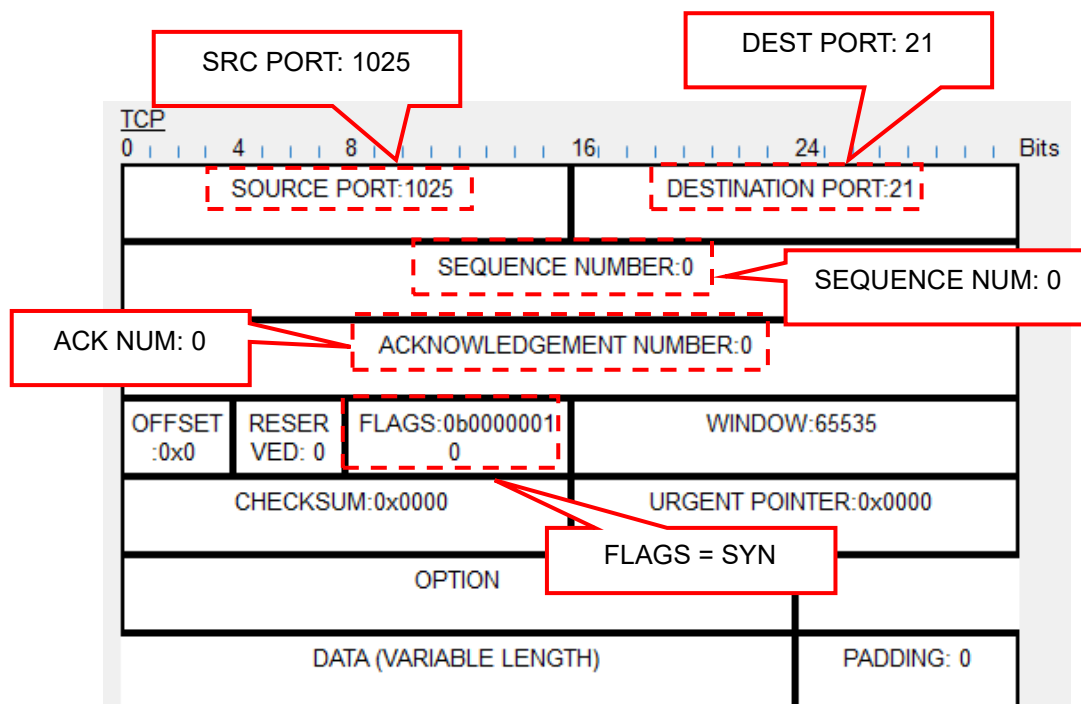
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?



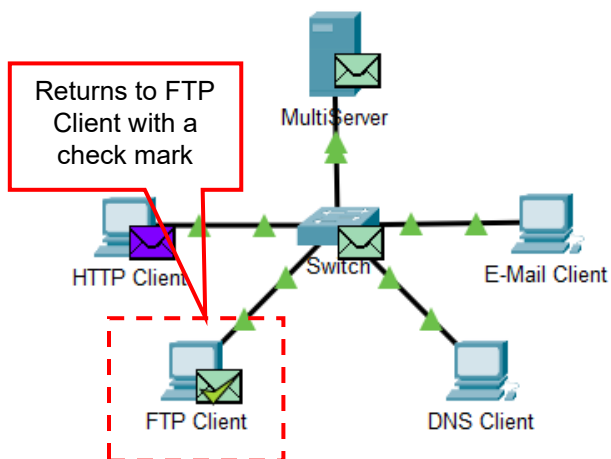
Are these communications considered to be reliable?

Yes. TCP is a reliable transport which has a connection oriented between sending and receiving process.

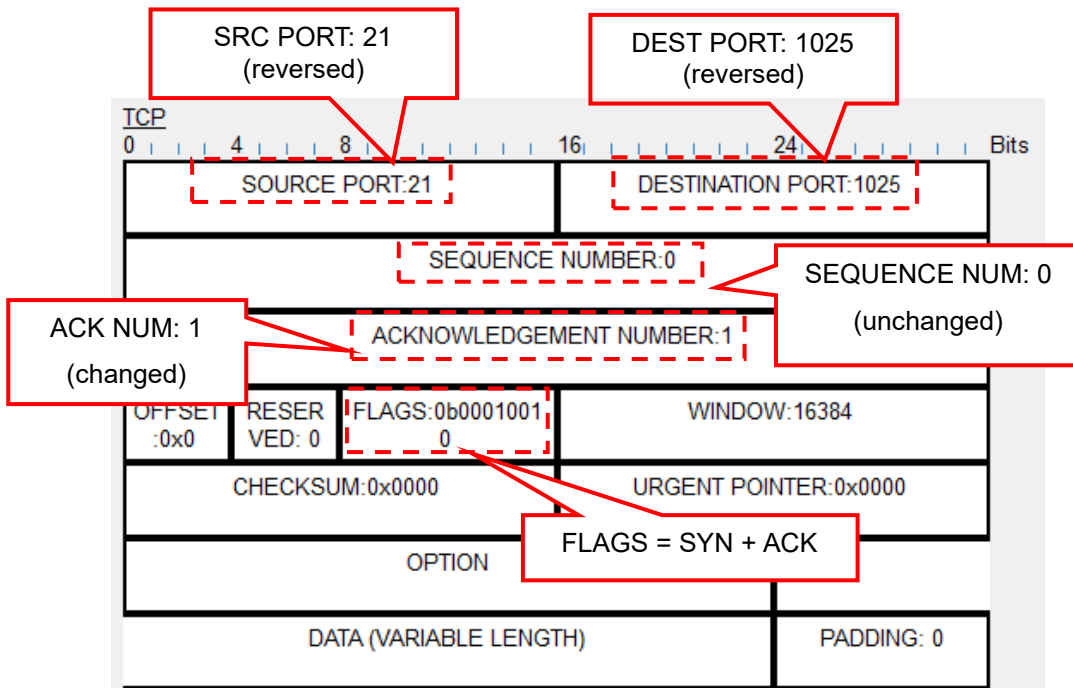
- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?



- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.

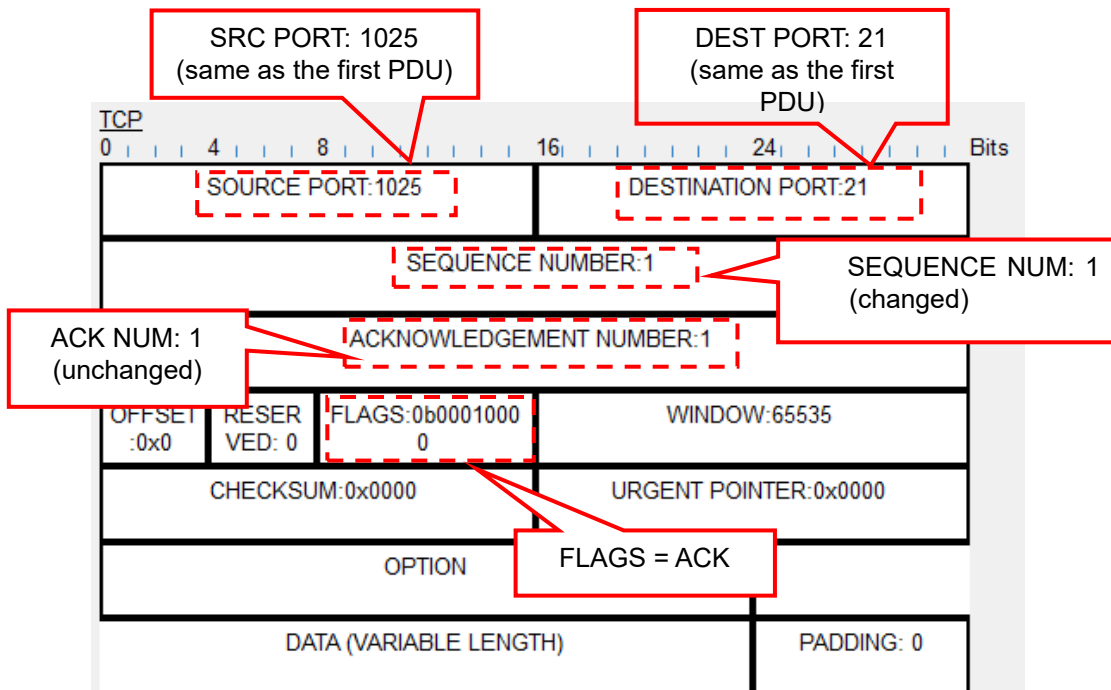


- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?



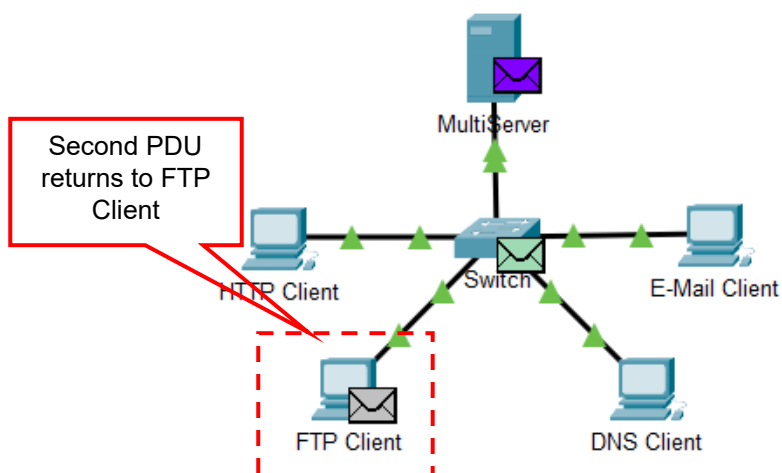
SRC PORT and DEST PORT are reversed, ACK changed to 1 and FLAGS changed to SYN + ACK

- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?



SRC PORT and DEST PORT are reversed, SEQUENCE NUM and ACK NUM changed to 1, and FLAGS changed to ACK

- h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.



- i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section. What is the message from the server?

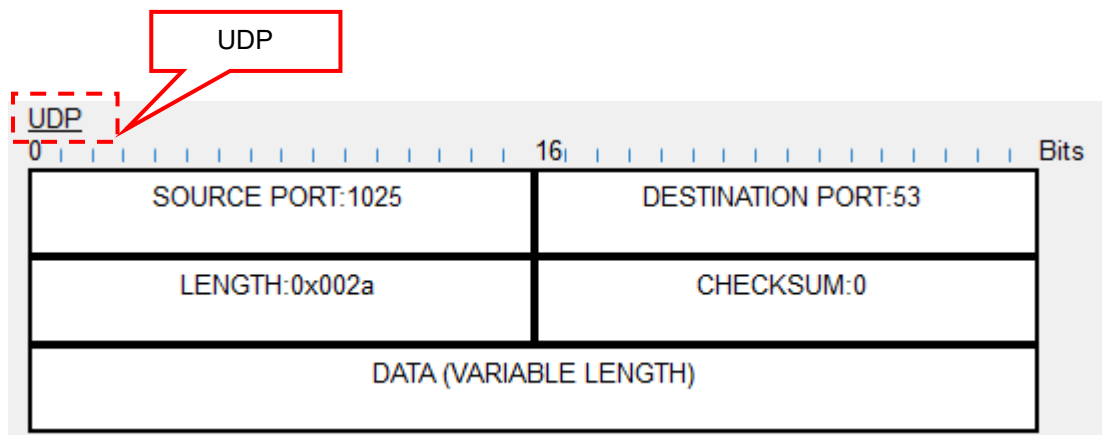
TCP															
0		4		8		16		24		Bits					
SOURCE PORT:21								DESTINATION PORT:1025							
SEQUENCE NUMBER:1															
ACKNOWLEDGEMENT NUMBER:1															
OFFSET: 0x0		RESERVED: 0		FLAGS:0b00011000				WINDOW:16384							
CHECKSUM:0x0000								URGENT POINTER:0x0000							
OPTION															
DATA (VARIABLE LENGTH)															
Message from the server: "Welcome to PT Ftp server"															

FTP Response															
0		4		8		16		Bytes							
Code:220															
Message:Welcome to PT Ftp server															

- j. Click **Back** until the simulation is reset.

Step 4: Examine DNS traffic as the clients communicate with the server.

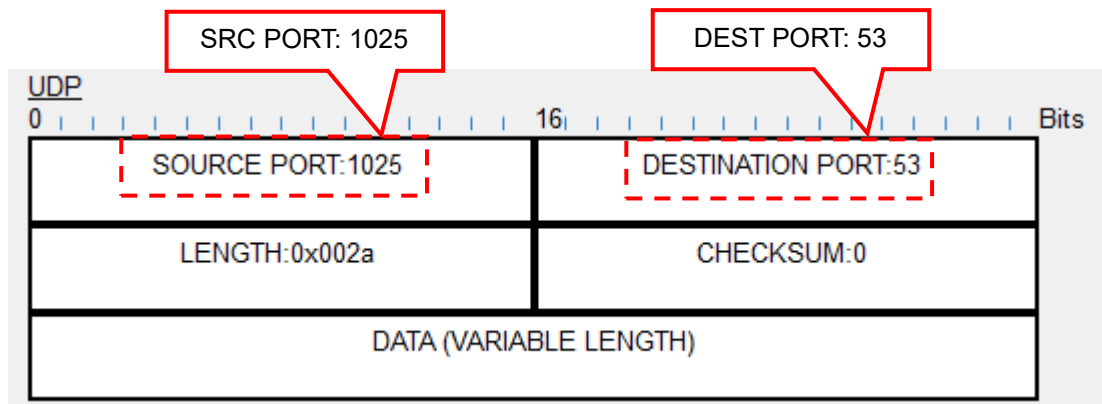
- In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.
- Click the PDU envelope to open it.
- Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?



Are these communications considered to be reliable?

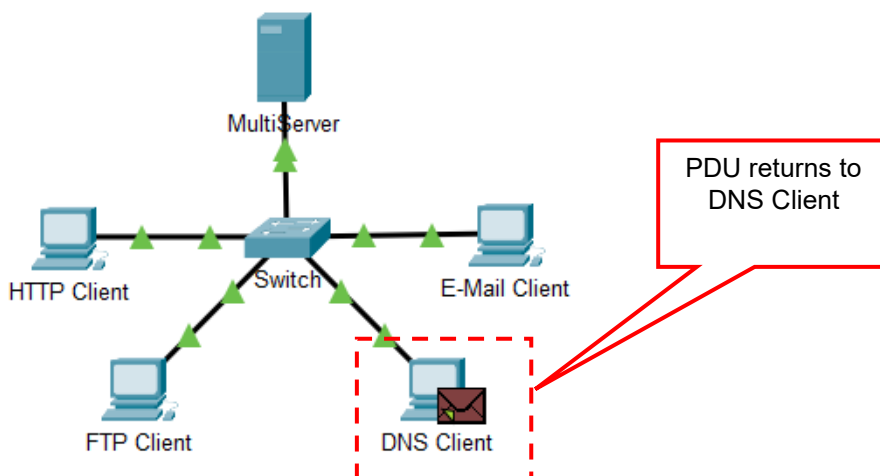
No. UDP is unreliable data transfer between sending and receiving process because it doesn't have a connection oriented.

- Record the **SRC PORT** and **DEST PORT** values. Why is there no sequence and acknowledgement number?

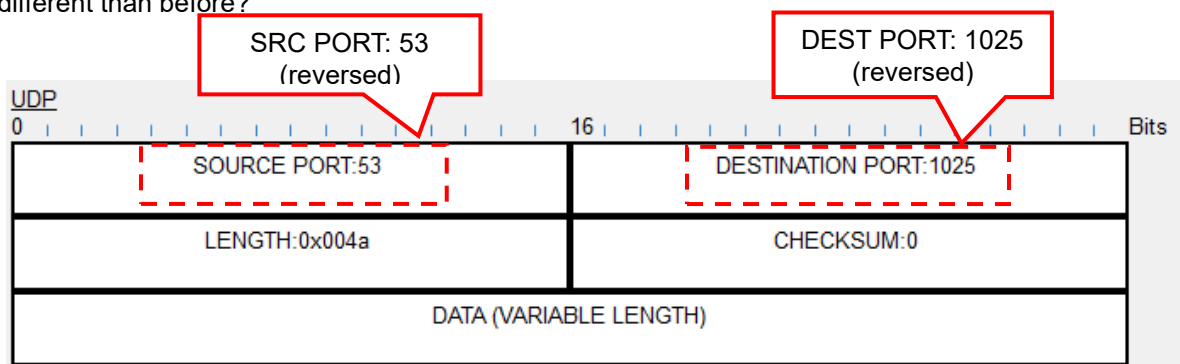


There is no SEQUENCE NUM and ACK NUM because UDP doesn't need to establish a reliable connection.

- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **DNS Client** with a checkmark.

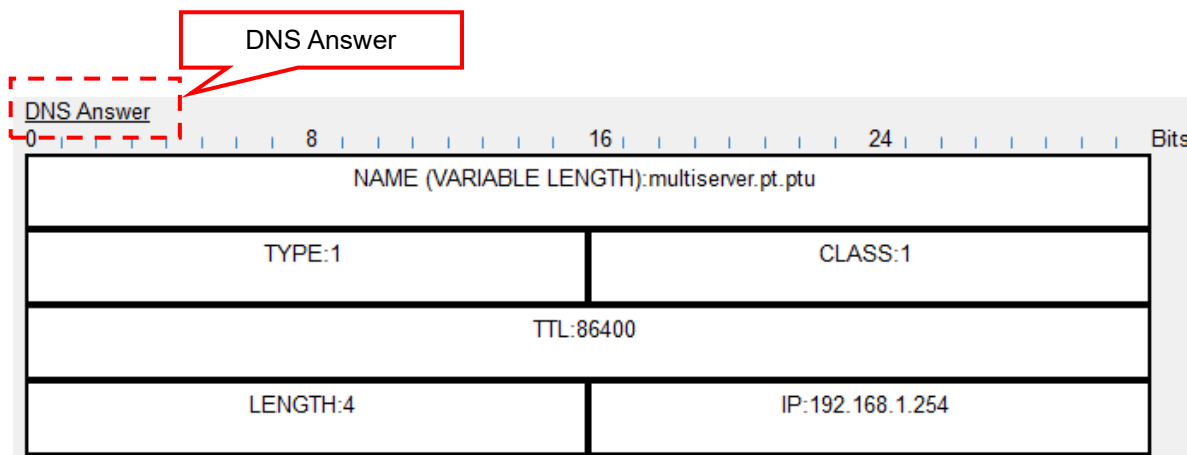


- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?



There is not any SEQUENCE NUM and ACK NUM show in UDP connection.

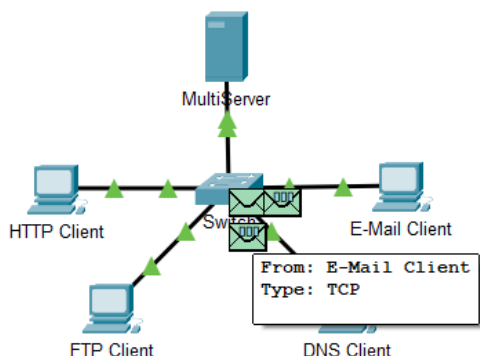
- g. What is the last section of the **PDU** called?



- h. Click **Back** until the simulation is reset.

Step 5: Examine email traffic as the clients communicate with the server.

- In the Simulation Panel, change **Edit Filters** to display only **POP3, SMTP** and **TCP**.
- Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **E-mail Client**. Click that PDU envelope to open it.



- Click the **Inbound PDU Details** tab and scroll down to the last section. What transport layer protocol does email traffic use?

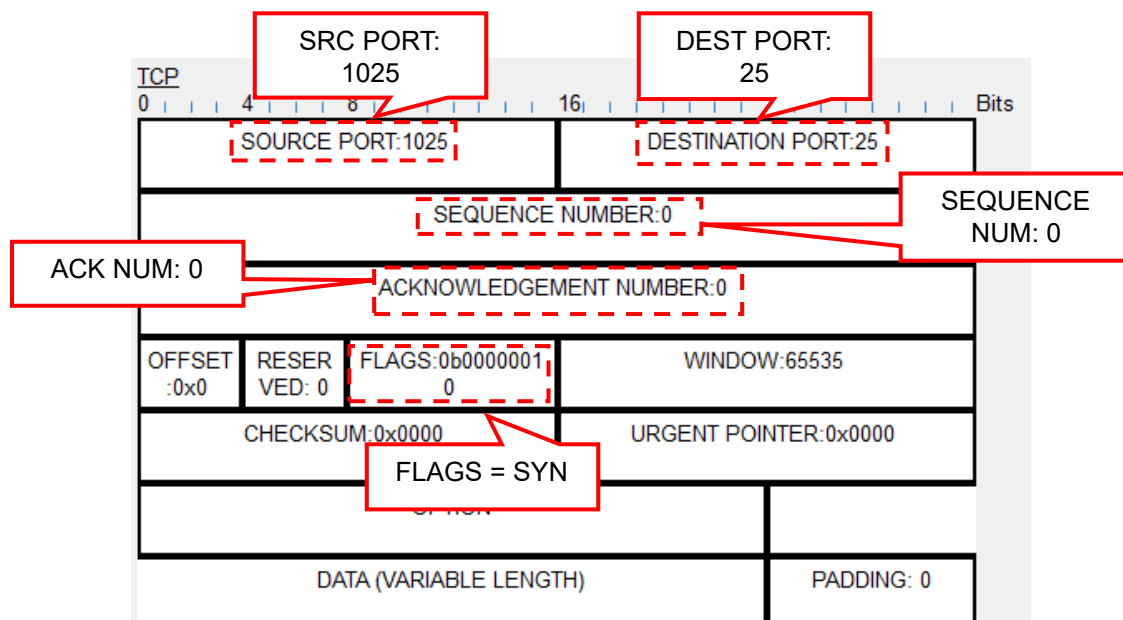
email traffic use?

TCP																			
0				4				8				24				Bits			
SOURCE PORT:102								DESTINATION PORT:25											
SEQUENCE NUMBER:0																			
ACKNOWLEDGEMENT NUMBER:0																			
OFFSET :0x0				RESER VED: 0				FLAGS:0b0000001 0				WINDOW:65535							
CHECKSUM:0x0000								URGENT POINTER:0x0000											
OPTION																			
DATA (VARIABLE LENGTH)												PADDING: 0							

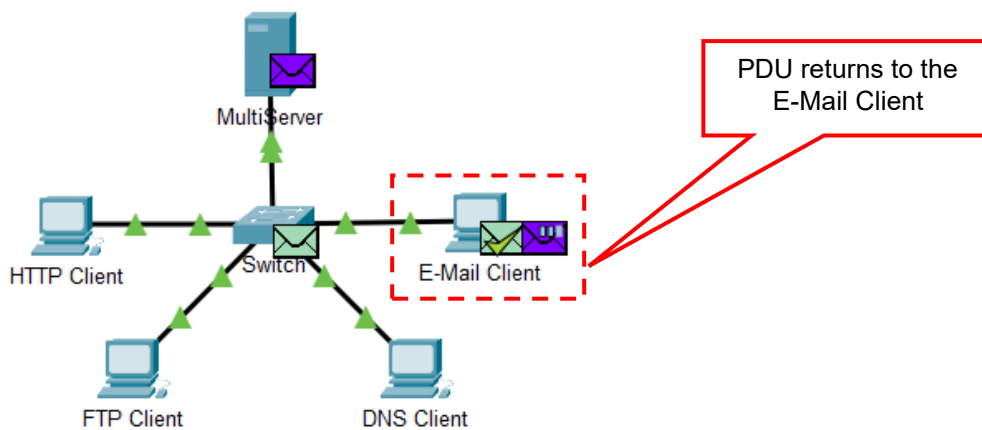
Are these communications considered to be reliable?

Yes. TCP is a reliable transport which has a connection oriented between sending and receiving process.

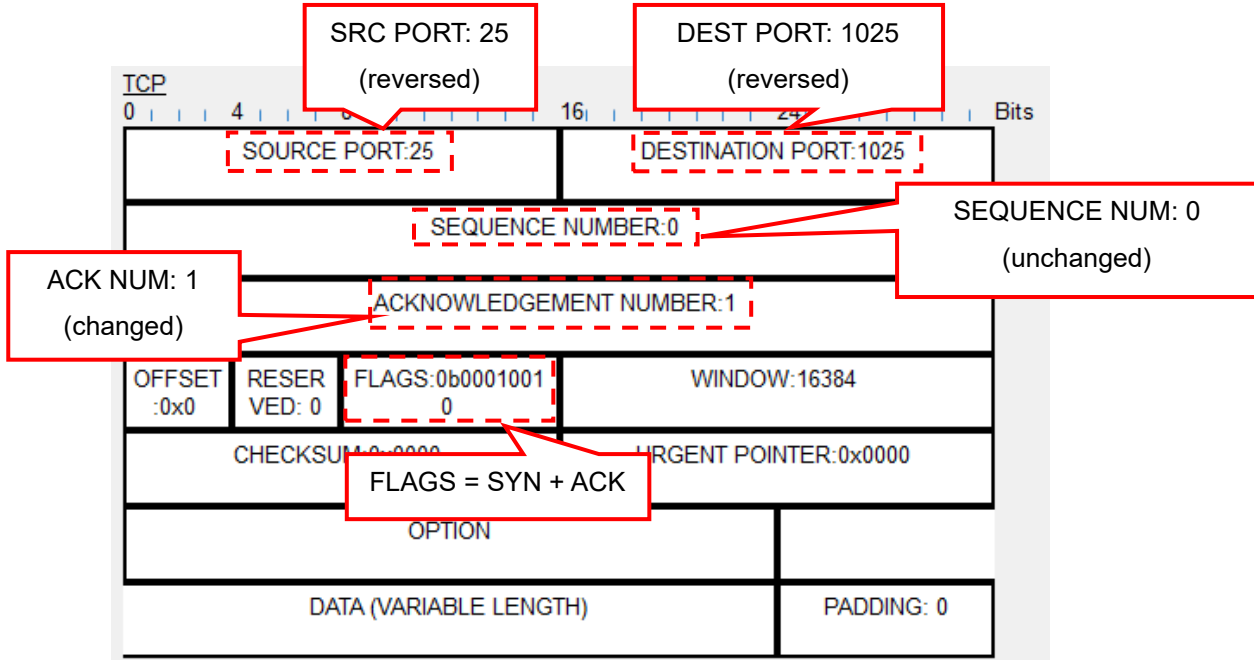
- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?



- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.

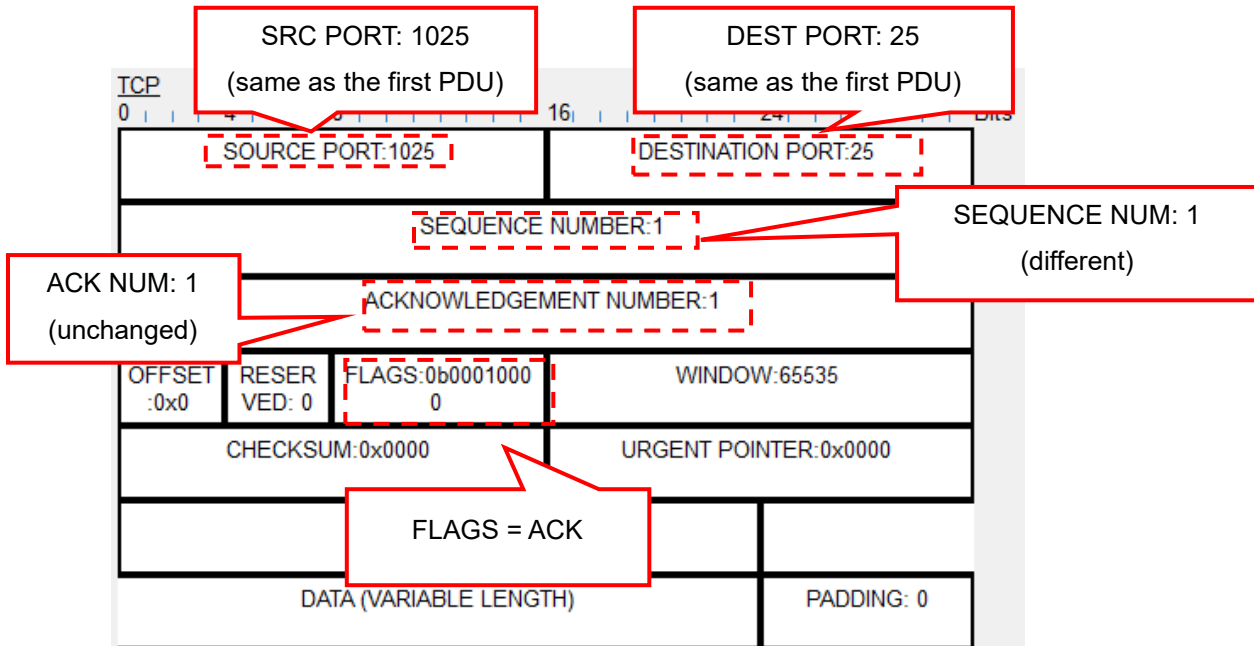


- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?



SRC PORT and DEST PORT are reversed, ACK NUM changed to 1 and FLAGS changed to SYN + ACK

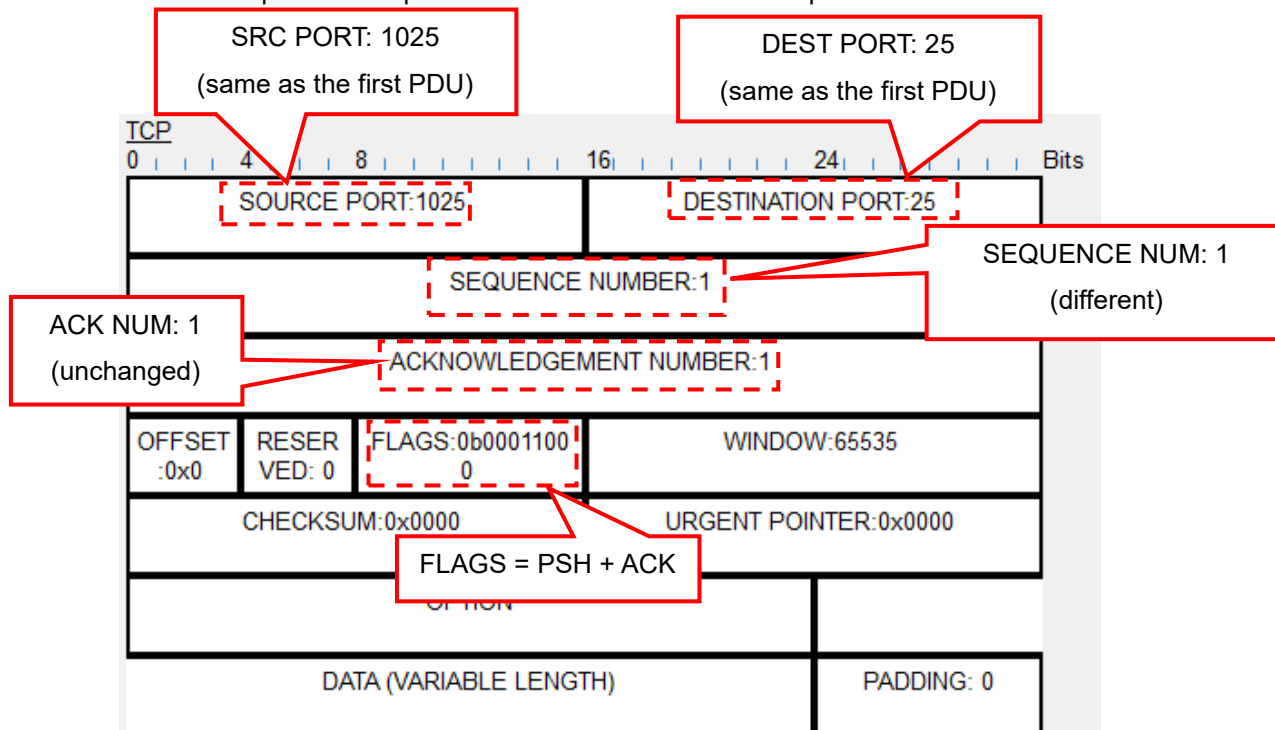
- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?



SRC PORT and DEST PORT are reversed, SEQUENCE NUM and ACK NUM changed to 1, and FLAGS changed to ACK

h. There is a second **PDU** of a different color that **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.

i. How are the port and sequence numbers different from the previous two **PDU**s?



SRC PORT and DEST PORT are reversed, SEQUENCE NUM and ACK NUM changed to 1, and FLAGS changed to PSH + ACK

j. What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

SMTP is associated with TCP port 25 and POP3 is associated with TCP port 110

k. Click **Back** until the simulation is reset.

Step 6: Examine the use of port numbers from the server.

- To see TCP active sessions, perform the following steps in quick succession:
 - Switch back to **Realtime** mode.
 - Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- Enter the **netstat** command. What protocols are listed in the left column?


```
C:\>netstat
```

Proto	Local Address	Foreign Address	State
TCP	192.168.1.254:21	192.168.1.2:1025	ESTABLISHED

C:\>

What port numbers are being used by the server?

```
C:\>netstat
```

Proto	Local Address	Foreign Address	State
TCP	192.168.1.254:21	192.168.1.2:1025	ESTABLISHED

C:\>

- c. What states are the sessions in?

```
C:\>netstat
```

Proto	Local Address	Foreign Address	State
TCP	192.168.1.254:21	192.168.1.2:1025	ESTABLISHED

C:\>

- d. Repeat the **netstat** command several times until you see only one session still ESTABLISHED. For which service is this connection still open?

```
C:\>netstat
```

Proto	Local Address	Foreign Address	State
TCP	192.168.1.254:21	192.168.1.2:1025	ESTABLISHED

C:\>netstat

Proto	Local Address	Foreign Address	State
TCP	192.168.1.254:21	192.168.1.2:1025	ESTABLISHED

C:\>netstat

Proto	Local Address	Foreign Address	State
TCP	192.168.1.254:21	192.168.1.2:1025	ESTABLISHED

C:\>

Why doesn't this session close like the other three? (Hint: Check the minimized clients)

```

Cisco Packet Tracer 6.0.1
C:\>ftp 192.168.1.254
Trying to connect...
Connected to 192.168.1.254
220- Welcome to PT Ftp server
Username:
  
```

The server is waiting for the user input

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 2: Examine Functionality of the TCP and UDP Protocols	Step 1	15	
	Step 2	15	
	Step 3	15	
	Step 4	15	
	Step 5	15	
	Step 6	25	
Total Score		100	